GlucoTwin clinic strategy

An auditable patient model connected to staffed review, documented follow-up and observed outcomes.

Team alpha | vnrvjiet | Shanmukh sai gara | registered size 1

Recommendation

Continue GlucoTwin. Concentrate four workstreams on a complete clinic journey and stronger predictive evidence. The product serves adult Type 2 diabetes care teams during defined monitoring windows.

Seven product capabilities

- Patient review queue with risk, freshness, ownership and review status.
- Patient evidence timeline with historical context, CGM and source labels.
- 30-minute forecasting, separate 60-minute results and visible adaptation.
- Staffed monitoring controls and outside-hours status.
- Persistent reviewed / follow-up / closed actions and audit records.
- Consultation summary PDF with evidence, gaps and review notes.
- Anonymized CSV, simulated streaming and limited local FHIR intake.

Current claim boundary

The implementation and internal retrospective evaluation can be checked locally. Doctor feedback, clinical effectiveness, live-device access and Indian clinical generalization are not established. Public demonstrations use anonymized or synthetic profiles.

Candidate comparison

Qualitative decision under public-data, software-only and four-workstream constraints. This is not an official scoring rubric.

Rank / idea	Decisive evidence and tradeoff
1 / GlucoTwin	Matched clinical context and native CGM; observable future outcomes and clear clinic demonstration. Existing twins make originality dependent on evidence and workflow.
2 / DialysisTwin	Strong longitudinal dialysis/BP data. Irregular measurements and treatment changes complicate warnings.
3 / SleepTwin	UCDDB overnight signals and labels. Small laboratory cohort; distinguish prediction from detection.
4 / RhythmTwin	Pre-AF recordings exist; selected pairs and limited historical context weaken continuous deployment claims.
5 / PressureTwin	Aurora-BP centers on cuffless measurement; future-event validity and redistribution restrictions remain issues.
6 / CardioTwin	Clinically compelling; relevant individual HF data access is a material deadline dependency.
7 / BreathTwin	Relevant longitudinal wearable work exists; suitable immediately downloadable event data were not verified.

Why retain GlucoTwin

It combines accessible linked data, direct evaluation against subsequent readings, visible patient updates and an understandable doctor workflow. Stronger personalization and event prediction matter more than adding more screens.

Data and methodological risks

Use observations at their true sampling resolution and preserve each source's evidence boundary.

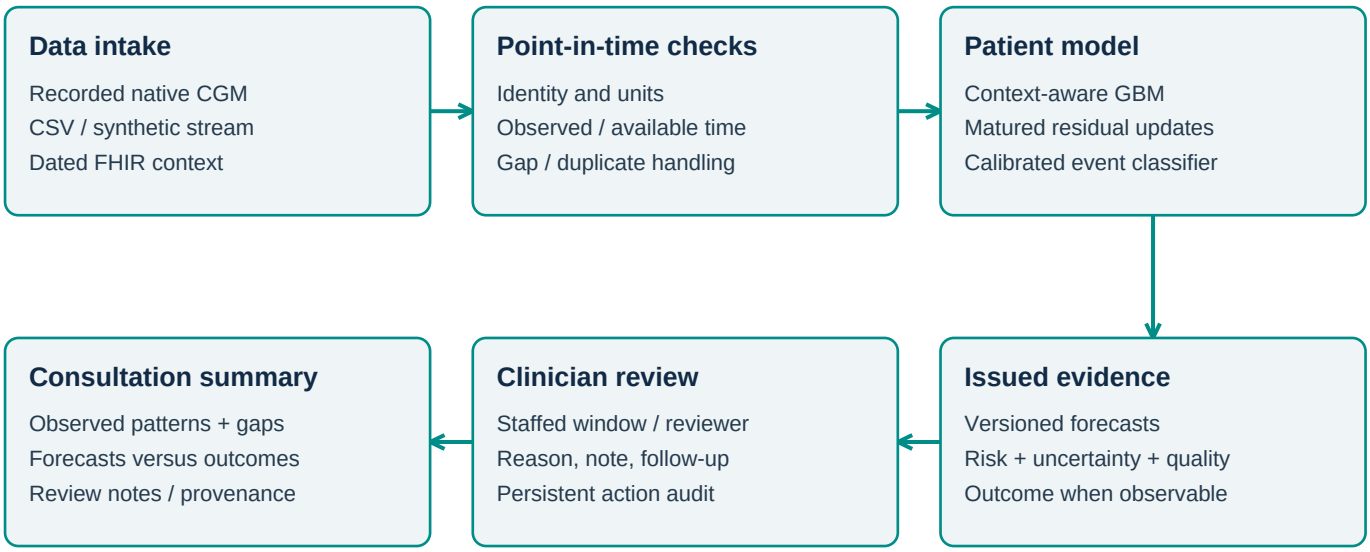
Item	Decision
Shanghai source	100 patients, 109 recordings, 112,462 usable measured CGM observations; native 15-minute sampling; matched historical context.
Data quality	13 missing rows excluded; 11 native gaps retained. No future imputation or interpolated observations.
Context timing	Age, sex, BMI and duration use documented historical-availability assumptions. Untimed labs and medications remain excluded.
New intake	Require identity, units, observation time and arrival/availability time. Idempotent duplicate handling and late-data rules.
Synthetic EHR	Dated synthetic scenarios demonstrate ingestion separately from the real measured-data benchmark.
CGMacros	Separate supplementary cohort only. One-minute interpolated exports are not new sensor observations; native timing and license constraints must be respected.
Generalization	Shanghai retrospective results do not establish ambulatory Indian clinic performance.

Source attribution

Code remains MIT; Shanghai source data retain CC BY 4.0 attribution. CGMacros is outside this default release. Provenance is available through the app and data manifest.

Architecture and differentiation

The model learns from observable outcomes; the review workflow makes its evidence usable.



SQLite migrations persist observations, context, forecasts, reviews and audit records.

Authenticated clinic access; no model update from acknowledgement; future outcomes stay hidden.

Model contribution

The upgraded candidate combines a context-aware population forecaster with bounded patient residual adaptation. Corrections update only when earlier forecast targets mature. A dedicated classifier matches the any-excursion event label.

Practical contribution

Persist issued evidence, review reasons, follow-up and action history. Preserve data freshness, uncertainty and unsuccessful predictions. This gives clinicians an inspectable process instead of an isolated health score.

Evaluation and acceptance

Frozen patient-separated internal validation. Earlier inspected patients are not called untouched validation.

- Five outer patient folds; all recordings from a patient stay together.
- Grouped inner parameter selection and separate calibration patients.
- Population versus identical model with residual adaptation; context-free, persistence and trend comparisons.
- MAE/RMSE; episode precision/recall; warning lead time; false alarms; Brier score; interval coverage and abstention.
- Patient bootstrap uncertainty; cold-start/adapted analysis; explicit failures.

Primary-model decision

30-minute evidence	Measured result
Population / hybrid MAE	9.812 / 9.606 mg/dL
Paired MAE difference / 95% CI	-0.206 / [-0.336, -0.110] mg/dL
Episode precision / overall recall	62.96% / 76.06%
False alerts / median lead	0.919 per 24 eligible hours / 15 min
90% interval coverage	89.68%

A small internal adaptation gain supports the hybrid primary. Historical-context benefit was inconclusive. Alert rates use all eligible forecast time, not staffed hours; the doctor-reviewed operating point is pending.

Functional acceptance

Future-data invariance, matured-only adaptation, identity isolation, duplicate suppression, missing-data abstention, persistent review, protected access and matching PDF exports. Rehearse the exact judge journey three times from a fresh startup.

Four workstreams and calendar

Parallel ownership, daily integration and a protected submission buffer.

Owner	Deliverable
A / Data and evaluation	Frozen protocol, grouped evaluation, calibration, paired comparisons, failures.
B / Twin and backend	Forecasting integration, ingestion, SQLite migrations, authenticated review workflow, reports.
C / Product and design	Queue, evidence timeline, review actions, monitoring controls, intake, report download.
D / Review and submission	Doctor packet, evidence boundaries, integration, architecture, presentation, video, submission checks.

Milestones

Dates	Target
10-11 October	Workflow review requested; engineering protocol and interfaces frozen.
12-14 October	Complete ingestion-to-forecast-to-review-to-report journey.
15-16 October	Evaluation, second review, persistence and failure handling.
17 October	Feature freeze; defensible primary model; three demo passes.
18-19 October	Documentation, presentation, architecture and recorded explanation.
20 October	Stop changes by 5 PM IST; deadline 7 PM IST.

Demonstration and submission

One patient, one warning episode, one documented review and one observed outcome.

Three-minute core journey

Open review queue; inspect patient; show advance warning and adaptation; save a clinician review; reveal the observed outcome; demonstrate sensor-gap suppression; export the consultation summary. Identify recorded playback and retrospectively selected cases.

Explanation recording

The longer walkthrough covers the problem, architecture, actual app, evaluation, failures, reproducibility and future pilot gates. Synthetic narration is disclosed; doctor review is never implied unless completed.

Submission package

Public repository, README with team/college details, problem/use case, stack and model details; open-source license; architecture and presentation PDF/PPT; explanation/demo video of at least 20 minutes. Registration: 18 October, 7 PM IST. Prototype: 20 October, 7 PM IST.

Human inputs still required

Team alpha, vnrvjiet and Shanmukh sai gara are supplied. The public source and assets are published at github.com/shanmukhgara48-source/glucotwin-alpha. Complete actual doctor feedback, registration and platform submission using the participant account; no completed entry is implied.

Pilot gates

Before live clinical use: intended-use and staffing protocol, protected deployment, authorized device timing, applicable privacy/regulatory/ethics assessment, prospective shadow evaluation, and supervised release after agreed evidence requirements pass.

References and evidence boundaries

Clickable primary sources. Local benchmark artifacts are the source of model-performance claims.

01. Challenge brief and submission rules

02. Shanghai matched clinical/CGM dataset

03. Shanghai archive and attribution

04. CGMacros cohort, access and license

05. HEMOBP dialysis cohort

06. UCDDB sleep cohort

07. PAF prediction cohort

08. Aurora-BP agreement and scope

09. TRUE-HF study and data access

10. COPD study and access

11. Published ABDM FHIR R4 guide

12. Authorized Dexcom access

13. CDSCO Medical Device Software guidance

14. IMDRF clinical evaluation

15. Established diabetes twin research in India

16. Personalized data-assimilation research

Assumptions

Existing computer; software only; public/synthetic data; four workstreams; doctor available for requested feedback. No new patient-data access is assumed.

Evidence boundaries

Local implementation and retrospective internal evaluation are distinct from physician feedback, vendor authorization, clinical effectiveness and regulatory approval. Patient self-reports remain contextual until validated. No autonomous treatment recommendations.